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Does a time-weighted average price deter settlement manipulation? Evidence from Polymarket's five-minute bitcoin markets

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ABSTRACT

Contracts that settle on a market price invite trading that moves the price at settlement, and the standard remedy is a time-weighted average price. Whether the average removes the incentive is rarely observed. Polymarket's five-minute bitcoin contract paid on a single Chainlink price snapshot until 7 August 2026 and on a 30-second and then a 60-second average afterwards. We measure the settlement footprint in Binance spot data at 82,752 five-minute boundaries. Under snapshot settlement the final-ten-second volume share rose from 3.05 to 5.68 percent, tenfold volume spikes became six times as frequent, and near-the-strike prices reverted. The footprint vanished in the week of the switch. In the first three weeks of the 60-second average the share was within 0.03 percentage points of its pre-launch level, volume did not migrate to earlier bins, and near-the-strike moves no longer reverted. Liquidity rewards that Polymarket paid market makers for quoting the contract after the switch came to less than 0.2 percent of its turnover, and the footprint did not return when they were briefly withdrawn in mid-August.

1. Introduction

A contract that settles on a market price gives its holders a reason to trade that market at settlement. The incentive is old and general. It shows in the expiration-day volume of cash-settled index futures, in the last-minute purchases with which fund managers mark up quarter-end prices, in dealer trading around the currency fix and in option trading at the VIX settlement (Stoll and Whaley, 1987; Carhart et al., 2002; Evans, 2018; Griffin and Shams, 2018). Kumar and Seppi (1992) show why. A trader who holds the contract and moves the reference price at settlement gains more on the contract than the move costs in the reference market. The standard remedy is to settle on an average of many prices rather than on one print (Duffie and Stein, 2015; Duffie and Dworczak, 2021). How much the remedy achieves is rarely observed. Settlement rules change seldom, and when they change the reference market is rarely observed at the resolution at which manipulation happens.

Polymarket's five-minute bitcoin market provides such an observation. From 12 February 2026 the contract paid one dollar if the Chainlink BTC/USD price at the end of a five-minute window was at or above its value at the start, a rule we call snapshot settlement. Dai et al. (2026) document that Binance order flow spiked in the last seconds of each window, that prices reverted afterwards, and that 821 wallets earned about $8.2 million in pushed cycles. On 7 August 2026 Polymarket replaced the single price with a 30-second time-weighted average, and on 14 August with a 60-second average.[1] The contract, its horizon, its reference asset and its oracle provider did not change. Of the contract's terms, only the statistic computed from the Chainlink price changed. Polymarket also paid market makers liquidity rewards for quoting the averaged markets from the switch to 5 September, and Section 3.3 shows that they were small against the contract's turnover. This paper measures what happened to the footprint when the settlement statistic changed.

We build one observation per five-minute boundary from Binance 1-second klines for BTC, ETH, SOL and XRP between 1 July 2025 and 4 September 2026 and read each Polymarket cycle's settlement rule from its own resolution text. The main sample holds the 82,752 BTC boundaries at which only the five-minute contract settles. The outcome is the share of the window's volume that trades in its final ten seconds. Regime means and day-clustered regressions compare the 226 days

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[1] Polymarket's changelog, https://docs.polymarket.com/changelog/predictions, accessed 28 September 2026, records that from 7 August 2026 "crypto up/down markets now resolve using Chainlink-computed time-weighted average prices instead of a single price snapshot", that "both the price to beat and the final settlement price come from the applicable TWAP feed", with a 30-second lookback for five-minute markets, and that from 14 August "all 5-minute crypto markets now resolve using a 60-second Chainlink TWAP". The resolution text of each market names its data stream, data.chain.link/streams/btc-usd before the switch and btc-usd-twap-30s-streams or btc-usd-twap-60s-streams afterwards, and Section 3.1 reads the regime from that text. Polymarket's developers announced the change on 30 July 2026 in a post at https://x.com/PolymarketDevs/status/2082813706996772881.

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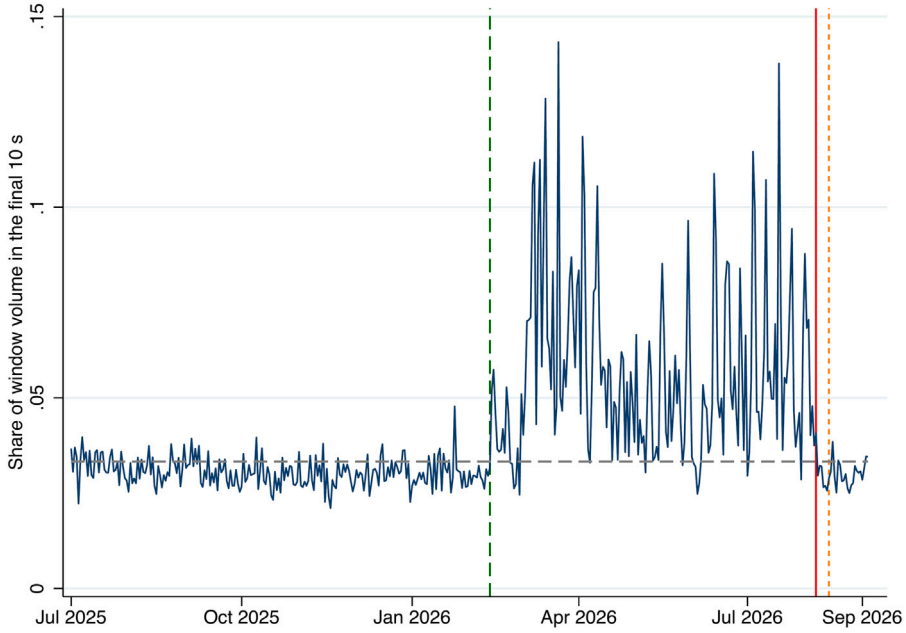


Fig. 1. Daily mean share of window volume traded in the final ten seconds before a boundary, BTCUSDT on Binance, boundaries at which only the five-minute contract settles, 1 July 2025 to 4 September 2026. The dashed horizontal line is the even-spread value of one thirtieth. Vertical lines mark the launch of the five-minute contract on 12 February 2026 (green, long dashes), the switch to a 30-second time-weighted average on 7 August (red, solid) and to a 60-second average on 14 August (orange, short dashes). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

before the launch, the 176 days of snapshot settlement and the 29 days of averaged settlement, and a weekly event study around 7 August dates the change. Online Appendix D also reports the results with the sample extended to 24 September.

The footprint appeared with the contract and vanished with the switch (Fig. 1). Before the launch the final-ten-second share was 3.05 percent. Under snapshot settlement it was 5.68 percent, the share of boundaries with a tenfold volume spike rose from 0.9 to 5.4 percent, and net taker flow in the final ten seconds more than tripled relative to the rest of the window. Under the 60-second average the share was 3.02 percent, within 0.03 percentage points of its pre-launch level, and the 95 percent interval rules out an excess above 0.11 points. Any surviving excess is at most a twentieth of the launch effect. The seven weekly coefficients before the switch are indistinguishable from the reference week, and the coefficient falls by 2.9 percentage points in the week the change took effect. In dollars, the final ten seconds carried $24 million a day more than an even spread implies under snapshot settlement and $5 million a day less under the 60-second rule, the same as before the launch.

The footprint was largest where a push could change the outcome. Under snapshot settlement, cycles whose Binance price one minute before the close sat within 5 basis points of the strike had a final-ten-second share of 7.46 percent, against 4.14 percent for cycles further from the strike. The share rose from 3.7 percent in the lowest quintile of contract volume to 8.5 percent in the highest. Near the strike a seventh of each final-ten-second price move reversed within 30 s. Under the 60-second rule the share lies between 2.7 and 3.2 percent in every quintile, the two moneyness groups sit at 3.14 and 2.90 percent, and near-the-strike moves no longer revert. Volume did not migrate to earlier parts of the window. No ten-second bin in the final minute differs from its pre-launch share by more than 0.30 percentage points. A placebo window in the middle of each cycle and the three assets with far smaller stakes show no comparable movement, and neither do the weeks around the taker-delay changes of 5 June and 17 August.

The contribution is an evaluation of a settlement-design change on the manipulation it was meant to stop, at one-second resolution in the reference market and with the contract held fixed. The closest precedent is the 2015 widening of the WM/R currency fix window from one to five minutes, which lengthened the window of a benchmark that already drew on many prices (Evans et al., 2018; Ito and Yamada, 2018). Here the operator moved from a single print to an average. Dai et al. (2026) identify the footprint and recommend longer contract horizons, and they name a settlement reference averaged over a window as a second lever. Polymarket kept the horizon and changed the statistic. Section 2 places the result in the literature on settlement manipulation and benchmark design. Section 3 describes the boundary panel and the before-and-after design, Section 4 reports the results, and Section 5 draws the implications for the design of price-settled contracts.

2. Related literature

Cash settlement moves the manipulation problem from the contract to the reference market. Kumar and Seppi (1992) show that an uninformed trader profits by taking a position in a cash-settled futures contract and then trading the spot asset at settlement, and Pirrong (2001) compares how vulnerable cash-settled and delivery-settled contracts are to a large trader with market power. A settlement value that depends on many prices is costlier to move than one that depends on a single print. Duffie and Dworczak (2021) formalise such a value as an estimator built from transaction prices, with weights that depend on transaction size, and show that its designer trades statistical efficiency, which calls for equal weights, against manipulation deterrence, which calls for less weight on small transactions. The length of the window over which the prices are collected is a separate lever, and Duffie and Stein (2015) discuss how benchmark definitions and fixing methods of this kind mitigate manipulation.

The footprint of settlement trading is well documented, but evidence on remedies is thinner. Ito and Yamada (2018) find that the volume spike in the fix window disappeared after the 2015 reform while the anomalies in price dynamics persisted, and Evans et al. (2018) find a benchmark that became harder to move and more representative of the day's prices at the cost of being harder for participants to replicate. In decentralised finance the same design choice arises for price oracles. Mackinga et al. (2022) show that the cost of moving an

on-chain time-weighted average had been overstated, because a single-block attack, or collusion with a block producer, moves it far more cheaply than the multi-block attack whose cost scales with the window.

Prediction markets add a second layer. A trader can move the contract's own price to influence beliefs or outcomes, which Hanson and Oprea (2009) model and Rasooly and Rozzi (2025) test with randomised price shocks in 817 markets. The contracts studied here instead settle on an asset price that holders can move in another market. Lee et al. (2026) observe that they are cash-or-nothing digital options on the reference price. Dai et al. (2026) model the resulting transfer from liquidity traders to manipulators and document the Binance footprint under snapshot settlement. Joshi and Zhou (2026) find price divergence between the oracle's constituent exchange and Binance around settlement for bitcoin contracts traded between February 2025 and January 2026, before the five-minute product existed. Gruener (2026) finds that the stated probabilities of the five-minute contracts are calibrated but not sharp. Gill (2026) studies settlement risk, and Wang (2026) shows that prices in disputed markets forecast the oracle's ruling rather than the event. None of these papers observes a change in the settlement rule. Two other design levers matter for reading the estimates. The liquidity rewards that came with the switch were a maker subsidy of the kind Malinova and Park (2015) study. Polymarket also changed its taker-delay rules on 5 June, 17 August and 4 September 2026, a lever that Perez (2026) analyses for in-play sports markets.

3. Empirical design

3.1. Data and sample

The reference market is Binance spot, the largest spot venue by volume and the one Dai et al. (2026) use. The Chainlink stream that settles the contract is computed from the prices of several venues, and Chainlink does not publish which venues enter it or how they are weighted. Polymarket's Gamma API reports both Chainlink values, the price to beat and the settlement value, for 30,835 cycles of the sample, nearly all between 22 March and 4 September 2026. As Online Appendix C shows, in these cycles the Binance return, measured as each rule prescribes, has a correlation of 0.996 to 0.999 with the Chainlink return and has the sign of the settled outcome in 95 to 97 percent of cycles. Binance is therefore an observable and close proxy for the reference price rather than the reference price itself. We use the public 1-second kline archive for BTCUSDT, ETHUSDT, SOLUSDT and XRPUSDT from 1 July 2025 to 4 September 2026, the last complete day in the archive when we built the main sample. Each second carries the open and close price, quote volume, the number of trades and the taker-buy quote volume, and every second in the 431 days is present. The Polymarket side comes from the Gamma API, which returns for each five-minute up-or-down market of the four assets its start and end time, traded volume, realised outcome and resolution text. The text dates the regimes to the minute. The last snapshot cycle ended at 00:00 UTC on 7 August, the 30-second rule ran until 00:00 UTC on 14 August, and the 60-second rule applies from 00:05 UTC on 14 August. Under the averaged rules both the price to beat and the settlement price are values of the averaged feed. Polymarket announced the change on 30 July 2026, eight days before it took effect, together with up to $1 million of liquidity rewards for the affected markets during August. The reward payments in Online Appendix B show that the reward programme ran from the first averaged cycle to 5 September, so the averaged period of the main sample falls within it. The BTC contract first traded on 12 February 2026 and the ETH, SOL and XRP contracts on 18 February 2026.

The unit of observation is an asset and a five-minute boundary T, with the window $[T-300, T)$. A boundary at :05, :10, :20, :25, :35, :40, :50 or :55 min past the hour settles only the five-minute contract. Boundaries at :15, :30 and :45 also settle the fifteen-minute contract, and the hourly boundary settles longer horizons as well. The main sample holds the 82,752 BTC boundaries of the first type. Of these, 43,392 precede the launch, 33,792 fall under snapshot settlement, 1344 under the 30-second rule and 4224 under the 60-second rule. Traded BTC cycles carried 91,600 shares of contract volume on average under snapshot settlement and 54,100 under the 60-second rule, against 9100 for ETH, 3800 for SOL and 2400 for XRP. Gamma reports volume in shares, each paying one dollar if its outcome occurs, and a BTC share traded at about 50 cents on average. Gamma reports no volume for most BTC cycles between 5 and 26 March 2026 although the Binance footprint continued, so results that use contract volume exclude those days and any cycle with a missing volume field.

3.2. Variable construction

Let v_b denote the quote volume in ten-second bin $b = 0, \ldots, 29$ of a window. The main outcome is the final-ten-second share $s_{29} = v_{29} / \sum_b v_b$. A spike indicator equals one when the volume rate in the final ten seconds exceeds ten times the rate over the preceding 290 s. The net-flow ratio divides the absolute taker net flow in the final ten seconds, twice the taker-buy volume minus total volume, per second by the volume per second over the rest of the window. It captures one-sided pressure rather than volume alone. The final-ten-second return is the log change from the close at $T-11$ to the close at $T-1$ in basis points, and post-boundary returns run from the close at $T-1$ to the close 30, 60 and 120 s later. Moneyness is proxied by the log distance between the Binance close at $T-61$ and a Binance strike that follows the contract's rule, the rule-matched strike, measured before the settlement-price averaging window opens. The Binance strike is the open at $T-300$ under snapshot settlement. Under the averaged rules it is the mean 1-second close over the 30 or 60 s before $T-300$. As Online Appendix C shows, within a day the gap between the Binance strike and the Chainlink price to beat varies with a standard deviation of about one basis point. A cycle is near the strike when this distance is within 5 basis points, which holds for about half of all cycles.[2] Realised variance over the first 240 s is a pre-determined volatility control. A placebo window ends 150 s before each boundary and has its own final-ten-second share.

3.3. Design

The regime of a boundary is the rule written in the cycle that settles at it, and boundaries before the first trading date form the pre-launch base. We estimate

$$y_t = \alpha + \beta_1 \,\text{snapshot}_t + \beta_2 \,\text{TWAP30}_t + \beta_3 \,\text{TWAP60}_t + \varepsilon_t, \tag{1}$$

[2] The strike is not a number that Polymarket chooses. It is the value of the Chainlink BTC/USD data stream at the moment the window opens. Chainlink is an oracle network that computes one price from the data of several exchanges. It does not publish which venues enter the stream, how they are weighted or how the averaged values are sampled, and Polymarket's documentation at https://docs.polymarket.com/market-data/chainlink-twap, accessed 28 September 2026, notes that the 30- and 60-second values are lookback windows rather than publication cadences. Each market's resolution text names the stream it settles on. Under snapshot settlement the contract compared the feed's value at the close with its value at the open. Under the averaged rules the named stream reports a 30- or 60-second time-weighted average of the underlying price, so the strike is the average over the seconds before the open and the settlement price is the average over the seconds before the close. Our proxy replaces the Chainlink strike with a Binance strike computed by the same rule and measures the distance one minute before the close. The Binance strike differs from the Chainlink strike by the aggregation and timing differences between one exchange and the feed. Because the proxy is measured at $T-61$, before the settlement-price averaging window opens, trading inside the settlement window cannot move it.

Table 1
Regime means and effects.

	Final-10 s share	Spike (> 10)	Net-flow ratio	|Push| (bp)	Final-60 s share
Panel A. Means by regime					
Pre-launch (226 days, 43,392)	0.0305	0.0091	0.7492	1.0491	0.1906
Snapshot (176 days, 33,792)	0.0568	0.0545	2.5170	1.3211	0.2168
TWAP 30 s (7 days, 1,344)	0.0305	0.0141	0.9476	0.4138	0.2271
TWAP 60 s (22 days, 4,224)	0.0302	0.0121	0.7924	0.9708	0.1921
Panel B. Regime coefficients, base = pre-launch					
Snapshot	**0.0262**	**0.0454**	**1.7678**	**0.2721**	**0.0262**
	(0.0018)	(0.0030)	(0.1771)	(0.0612)	(0.0022)
TWAP 30 s	0.0000	0.0051	0.1984	**−0.6352**	0.0365
	(0.0018)	(0.0047)	(0.1290)	(0.0854)	(0.0226)
TWAP 60 s	−0.0003	0.0030	0.0432	−0.0783	0.0015
	(0.0007)	(0.0023)	(0.1026)	(0.1246)	(0.0023)
95% CI, TWAP 60 s	[−0.0018, 0.0011]	[−0.0015, 0.0075]	[−0.1586, 0.2449]	[−0.3232, 0.1667]	[−0.0031, 0.0061]
95% CI, TWAP 60 s − snapshot	[−0.0303, −0.0228]	[−0.0496, −0.0351]	[−2.1251, −1.3242]	[−0.5946, −0.1060]	[−0.0307, −0.0187]
Panel C. Hour and day-of-week fixed effects and volatility control					
Snapshot	**0.0276**	**0.0476**	**1.8636**	**0.1736**	**0.0301**
	(0.0017)	(0.0027)	(0.1654)	(0.0277)	(0.0024)
TWAP 30 s	**−0.0100**	**−0.0112**	**−0.5189**	0.0940	0.0076
	(0.0031)	(0.0043)	(0.2165)	(0.0816)	(0.0174)
TWAP 60 s	−0.0017	0.0008	−0.0578	0.0066	−0.0023
	(0.0019)	(0.0024)	(0.1409)	(0.0459)	(0.0041)

Notes: BTC five-minute-only boundaries (N = 82,752). Panel A reports regime means; parentheses give days and boundary counts. Panel B reports Eq. (1), with pre-launch as the base, day-clustered standard errors in parentheses, and 95% confidence intervals. Panel C adds hour and day-of-week fixed effects and log first-240-second realised variance. Shares are fractions of window volume. Spike denotes a final-10 s volume rate above ten times the preceding rate; the net-flow ratio measures absolute net taker pressure; |Push| is the absolute final-10 s log return in basis points. Coefficients in bold are statistically significant at the 5% level ($p<0.05$).

with standard errors clustered by day. The launch effect is β_1, the level under the final rule relative to a period without any five-minute contract is β_3, and $\beta_3 - \beta_1$ is the change at the switch. A second specification adds hour-of-day and day-of-week fixed effects and the realised variance of the first 240 s. A weekly event study on daily means around 7 August, with week −1 omitted and only full seven-day weeks kept, dates the change. The remaining tests change the outcome rather than the specification. They cover the volume profile across all thirty bins, the interaction of the switch with moneyness, comparisons within fixed quintiles of contract volume, the slope of post-boundary returns on the final-ten-second return within each regime, and the same regressions on the placebo window, the shared boundaries, the other assets and the first two dates on which Polymarket changed its taker-delay rules. Table A1 in Appendix A defines every variable and sample restriction, and Online Appendix B dates each platform change.

The design compares periods and has no untreated market. It identifies the change in the footprint at the switch, and that change bundles the new settlement statistic with the liquidity rewards that began with the first averaged cycle and with any change in who traded the contract. The placebo window and the small-stake assets show no comparable movement in Binance volume at those times. They do not separate the reward programme from the rule change.

The reward payments measure the programme directly. Polymarket does not publish payouts by market, but each wallet's public activity record lists its reward payments, and Online Appendix B collects those of the 16,128 wallets that traded the contract from 7 to 31 August. They received $324,000 from all Polymarket markets, which is $45 per five-minute cycle and 0.17 percent of the contract's cash turnover of $188 million. Scaling each wallet's rewards by the share of its maker volume traded in this contract gives $160,000, or 0.085 percent of turnover. Passed entirely into quotes, the $324,000 would have paid makers 0.09 cents on each share traded, enough to narrow the bid–ask spread by 0.17 cents. Against the maker rebates implied by the fee schedule the rewards weighed more, 42 percent before allocation and 21 percent after it, and 71 percent of them went to wallets seen as makers. The programme was thus a subsidy to liquidity providers that was small against the money at stake in the contract. Its settings were withdrawn from 12 to 15 August. On 13 and 14 August the payments fell to their pre-programme level and the final-ten-second share was 2.6 and 2.9 percent, below its pre-launch level of 3.05 percent. The footprint therefore did not depend on the day's payments, although two days are too short to exclude a slower effect through who traded the contract.

4. Empirical results

4.1. The settlement footprint across regimes

Table 1 reports regime means and the estimates of Eq. (1). Before the launch the final-ten-second share was 3.05 percent, below the even-spread value of 3.33 percent because volume clusters at the open of each window. Under snapshot settlement it was 5.68 percent. The launch coefficient is 2.62 percentage points with a standard error of 0.18, an 86 percent increase over the base. Tenfold spikes occurred at 5.4 percent of boundaries, six times their pre-launch frequency, and the net-flow ratio rose from 0.75 to 2.52. Under the 60-second rule the share is 3.02 percent. Its coefficient relative to pre-launch is −0.03 percentage points with a 95 percent interval from −0.18 to +0.11. The difference from the snapshot period is −2.66 points with an interval from −3.03 to −2.28, which covers a full reversal of the launch effect. With hour and day-of-week fixed effects and the volatility control the 60-second coefficient is −0.17 points with a standard error of 0.19. The 30-second coefficient in that specification is −1.00 points with a standard error of 0.31, because realised variance in that week was a fifth of the sample average and the control carries a negative coefficient.[3] In dollars, the final ten seconds carried $24 million a day more than an even spread implies under snapshot settlement, against $5 million a day less both before the launch and under the 60-second rule.

[3] The volatility control drives the change. In the share regression the log realised variance of the first 240 s enters with a coefficient of −0.0070 and a standard error of 0.0006, because windows with more early volatility spread their volume more evenly, and its mean is 2.16 in the 30-second week against 3.60 before the launch. The unusually calm week therefore predicts a final-ten-second share 1.00 percentage points above the pre-launch level, the observed share equals the pre-launch mean, and the gap is the Panel C coefficient of −1.00 points. The hour and day-of-week fixed effects account for almost none of the change.

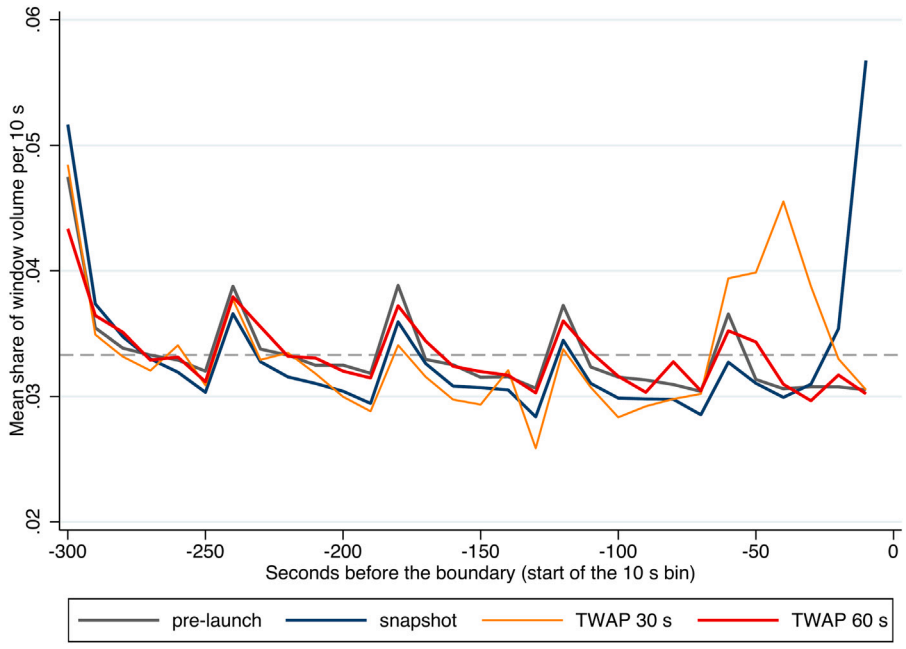


Fig. 2. Mean share of window volume in each ten-second bin of the five-minute window, by settlement regime, BTC five-minute-only boundaries. The horizontal axis gives the start of each bin in seconds before the boundary; the dashed line is the even-spread value. The spikes at −300, −240, −180, −120 and −60 s are the window open and the minute marks. The 30-second regime covers seven days, and its excess between −60 and −30 s comes from 8 and 9 August (Table 2, Panel C).

Fig. 1 shows the daily series. The level shifts up in the week of 12 February, stays elevated for six months and drops in the week of 7 August. Fig. 3 plots the weekly coefficients for the share and the spike frequency. The seven coefficients before the switch lie between −1.0 and +0.5 percentage points and none is distinguishable from zero. The four weeks after are −2.9, −2.9, −3.2 and −2.9 points, each with a standard error near 0.7. Across the full calendar months of snapshot settlement, March to July, monthly means range from 5.0 to 7.2 percent with no downward trend, and the seven days between the 30 July announcement and the switch averaged 6.0 percent. Traders did not withdraw before the rule took effect.

A 60-second average spreads weight over twice as many seconds of trading as a 30-second average, so a push of a given size moves it less. Under both rules the final-ten-second share sits at its pre-launch level, and the unadjusted 30-second and 60-second coefficients are indistinguishable, with $p = 0.87$. The controlled specification rejects their equality with $p = 0.019$, and the rejection comes from the volatility control, which predicts a higher share for the unusually calm 30-second week than was observed.

Online Appendix D reports the results with the sample extended to 24 September and the 60-second rule split at the end of the reward programme. From 14 August to 5 September, the programme period of the 60-second rule, the share averaged 3.10 percent, at its pre-launch level. It rose from Saturday 5 September, the day after Polymarket raised its taker delay, and from 6 to 24 September it averaged 4.02 percent, more than a third of the way from its pre-launch to its snapshot level, with the excess mainly on weekends, when Binance volume is thinnest. The near-the-strike reversal slopes of that period differ neither from zero nor from their snapshot values.

4.2. The intra-window volume profile

An average can be moved by trading throughout its window, so a falling final-ten-second share need not mean a vanished incentive. Fig. 2 and Table 2 follow the whole window. Under snapshot settlement the final bin gained 2.62 points and the penultimate bin 0.46 points, while bins 21 to 23 together lost 0.46 points and bins 24 to 26 lost 0.48 points. Volume moved from the body of the window into its last twenty seconds. Under the 60-second rule the change is small in every bin of the final minute. Bin 25, which covers the ten seconds from 50 to 40 s before the boundary, is 0.30 points above its pre-launch share of 3.14 percent, with a standard error of 0.09, and it is the only bin in the final minute that differs from its base at the 5 percent level. The other five bins differ by at most 0.13 points, and the final-minute share as a whole is 0.15 points above its base with a standard error of 0.23. The first bin of the window lost 0.41 points. The 30-second week is different. Its final-ten-second share sits at the pre-launch level, but bins 24 to 26, the thirty seconds before the averaging window opened, carry 2.6 points more than before the launch. The excess comes from two days. On Saturday 8 and Sunday 9 August the three bins carried 20 and 19 percent of window volume, against 9 to 10 percent on the other five days and 9.9 percent before the launch. Without the two days the 30-second coefficient on bins 24 to 26 is −0.20 points with a standard error of 0.22. No comparable concentration appears in the 22 days of the 60-second regime.

4.3. Where the footprint lived

A push pays when the cycle is close to its strike, so the footprint should sit in near-the-strike cycles, grow with the money at stake and be larger where more contracts settle. Table 3 shows all three. Under snapshot settlement, cycles near the strike had a final-ten-second share of 7.46 percent, a spike frequency of 8.4 percent and a net-flow ratio of 3.78, against 4.14 percent, 2.9 percent and 1.43 for cycles further away. Under the 60-second rule the two groups sit at 3.14 and 2.90 percent on the share, 1.6 and 0.8 percent on spikes and 0.95 and 0.63 on the ratio. The interaction of the pooled post-switch indicator with the near-the-strike indicator is −3.11 points on the share and −4.66 points on the spike frequency, with standard errors of 0.30 and 0.52. Pushed cycles, defined by a net-flow ratio above the snapshot-period 90th percentile, fell from 14.7 to 4.5 percent of near-the-strike cycles and from 6.0 to 3.2 percent of the others.

Across quintiles of contract volume, the snapshot-period share rises from 3.7 percent in the lowest quintile, where cycles carried 42,000

Table 2
Intra-window volume profile.

	Bin 0	Bins 21–23	Bins 24–26	Bins 27–28	Bin 29	Bins 24–29
Panel A. Grouped bins, all boundaries (N = 82,752)						
Snapshot	**0.0042**	**−0.0046**	**−0.0048**	**0.0048**	**0.0262**	**0.0262**
	(0.0007)	(0.0007)	(0.0007)	(0.0009)	(0.0018)	(0.0022)
TWAP 30 s	0.0010	−0.0034	0.0263	0.0102	0.0000	0.0365
	(0.0023)	(0.0019)	(0.0171)	(0.0054)	(0.0018)	(0.0226)
TWAP 60 s	**−0.0041**	0.0009	0.0020	−0.0002	−0.0003	0.0015
	(0.0010)	(0.0015)	(0.0018)	(0.0010)	(0.0007)	(0.0023)
Pre-launch mean	0.0475	0.0927	0.0985	0.0615	0.0305	0.1906
Panel B. Single bins of the final minute, all boundaries (N = 82,752)						
	Bin 24	Bin 25	Bin 26	Bin 27	Bin 28	Bin 29
Snapshot	**−0.0038**	−0.0003	−0.0007	0.0002	**0.0046**	**0.0262**
	(0.0004)	(0.0004)	(0.0004)	(0.0004)	(0.0007)	(0.0018)
TWAP 30 s	0.0028	0.0085	**0.0149**	0.0080	0.0022	0.0000
	(0.0044)	(0.0065)	(0.0071)	(0.0046)	(0.0014)	(0.0018)
TWAP 60 s	−0.0013	**0.0030**	0.0004	−0.0011	0.0009	−0.0003
	(0.0010)	(0.0009)	(0.0008)	(0.0006)	(0.0009)	(0.0007)
Pre-launch mean	0.0366	0.0314	0.0306	0.0308	0.0308	0.0305
Panel C. Excluding 8 and 9 August 2026 (N = 82,368)						
			Bins 24–26	Bins 27–28	Bin 29	Bins 24–29
TWAP 30 s			−0.0020	0.0020	−0.0001	−0.0001
			(0.0022)	(0.0027)	(0.0025)	(0.0068)
TWAP 60 s			0.0020	−0.0002	−0.0003	0.0015
			(0.0018)	(0.0010)	(0.0007)	(0.0023)

Notes: Regime coefficients relative to pre-launch, with day-clustered standard errors in parentheses. Panels A and B use all 82,752 BTC five-minute-only boundaries and report pre-launch means in their last rows. Bins 21–23 cover $[T-90, T-60)$, bins 24–26 cover $[T-60, T-30)$, and bins 27–28 cover $[T-30, T-10)$. Bin 0 is the first ten seconds and bin 29 the last ten seconds. Panel B reports the six single bins of the final minute; bin b covers $[T-300+10b,\ T-290+10b)$. Panel C excludes 8–9 August 2026 (N = 82,368). Coefficients in bold are statistically significant at the 5% level (p<0.05).

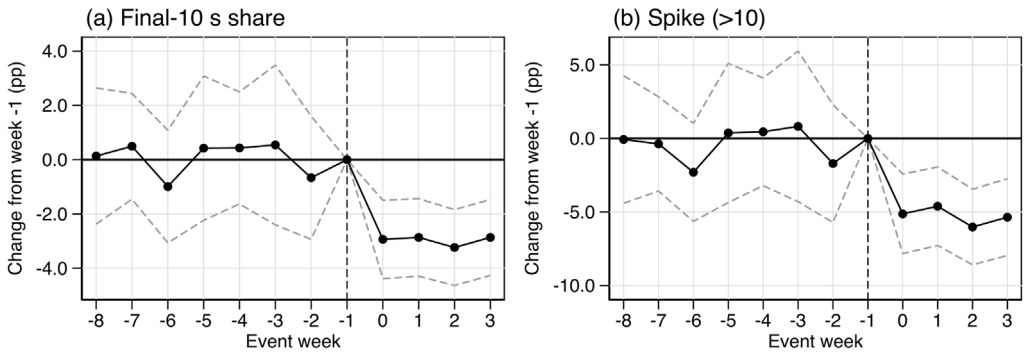


Fig. 3. Weekly event-study coefficients for (a) the daily mean final-ten-second volume share and (b) the frequency of tenfold volume spikes, BTC five-minute-only boundaries. Stata `eventdd` estimates the weekly dummy specification on 84 dates. Week 0 begins on 7 August 2026; week −1 is the omitted reference and is marked by the vertical dashed line. Its zero coefficient is a normalisation. Each week covers seven days. Black connected points are coefficients; grey dashed lines are pointwise 95 percent confidence limits based on heteroskedasticity-robust standard errors. The scale is percentage points. The four post-switch coefficients in each panel remain significant at the 1 percent level with Newey–West standard errors and three or seven daily lags (Newey and West, 1987).

shares, to 8.5 percent in the highest, where they carried 149,000. After the switch the share lies between 2.7 and 3.2 percent in every quintile. Contract volume fell by 37,900 shares per cycle after the switch, about 40 percent, so the two highest post-switch cells hold only 258 and 81 cycles. Contract volume and the mix of traders can respond to the switch, so the comparison locates the footprint but cannot separate the settlement rule from the fall in stakes. The launch effect also grows with the number of contracts that settle, from 2.62 points at five-minute-only boundaries to 3.49 points at the quarter-hour boundaries shared with the fifteen-minute contract and 4.81 points at the hour. All three boundary types are back at their pre-launch levels under the 60-second rule. The pattern is consistent with the finding of Dai et al. (2026) that the manipulation signature is much attenuated in the fifteen-minute contract itself. The five-minute contract settles at every shared boundary, and the fifteen-minute and hourly contracts add money at stake at the same second, which raises the payoff to a push that the five-minute contract already invites.

4.4. Price signature and placebos

Table 4 turns to prices. Within each regime we regress the return over the next 30, 60 and 120 s on the final-ten-second return. The pre-launch slopes lie between −0.01 and +0.08 and are indistinguishable from zero. Under snapshot settlement they are −0.07, −0.13 and −0.13, and in near-the-strike cycles −0.14, −0.18 and −0.16, each with a standard error between 0.02 and 0.05. Near the strike a seventh of a final-ten-second move reversed within 30 s and about a sixth within a minute. Under the 60-second rule the near-the-strike slope at 30 s is +0.30 with a standard error of 0.11, so late moves continue rather than revert over that horizon. The 60- and 120-second slopes, +0.24 and +0.06, have standard errors of 0.19 and 0.23, and the slopes away from the strike are close to zero. The change in the near-the-strike slope between the snapshot and 60-second regimes is significant at the 30- and 60-second horizons and not at 120 s, and none of the all-cycle slope changes is significant at the 5 percent level. The absolute

Table 3
Where the footprint lived.

	Final-10 s share		Spike (> 10)		Net-flow ratio	
Panel A. Moneyness	Near	Far	Near	Far	Near	Far
Snapshot	0.0746	0.0414	0.0840	0.0290	3.7814	1.4251
TWAP 60 s	0.0314	0.0290	0.0161	0.0081	0.9508	0.6331
Post × near	**−0.0311** (0.0030)		**−0.0466** (0.0052)		**−2.0399** (0.3379)	
Panel B. Recorded cycle volume, final-10 s share						
Quintile	1	2	3	4	5	
Mean snapshot volume (thousand shares)	42	62	78	99	149	
Snapshot	0.0368	0.0385	0.0436	0.0548	0.0851	
TWAP 60 s	0.0315	0.0298	0.0267	0.0297	0.0284	
Cycles, snapshot/TWAP 60 s	3885/2230	5614/988	6190/550	6562/258	6789/81	
Panel C. Share of pushed cycles						
	All	Near	Far			
Snapshot	0.1000	0.1468	0.0596			
TWAP 60 s	0.0388	0.0453	0.0323			
Adjusted difference	**−0.0613** (0.0069)	**−0.1015** (0.0097)	**−0.0269** (0.0044)			
Panel D. Boundary type, final-10 s share (all BTC boundaries)						
	Five-minute only	:15/:30/:45	:00			
Pre-launch mean	0.0305	0.0306	0.0364			
Snapshot	**0.0262** (0.0018)	**0.0349** (0.0027)	**0.0481** (0.0038)			
TWAP 60 s	−0.0003 (0.0007)	0.0005 (0.0016)	−0.0008 (0.0028)			
Boundaries	82,752	31,032	10,343			

Notes: Launched BTC five-minute-only boundaries (N = 39,360). Panel A reports means for the snapshot and 60-second regimes and the interaction of the post-switch indicator, equal to one under either averaged rule, with the near-the-strike indicator, with hour and day-of-week fixed effects. A cycle is near when the Binance close at $T-61$ lies within 5 bp of the rule-matched strike of Section 3.2. Panel B uses fixed cycle-volume quintiles pooled across launched regimes, excluding missing volume and 5–26 March (N = 34,476); the mean-volume row refers to snapshot observations, in shares that each pay one dollar if their outcome occurs. Panel C fixes the pushed threshold at the snapshot 90th percentile of the net-flow ratio; adjusted differences include hour and day-of-week effects. Panel D uses all BTC boundaries and pre-launch as the base. Parentheses contain day-clustered standard errors. Coefficients in bold are statistically significant at the 5% level (p<0.05).

Table 4
Price signature and placebos.

	All cycles			Near the strike		
Panel A. Slope of the post-boundary return on the final-10 s return						
Horizon	30 s	60 s	120 s	30 s	60 s	120 s
Pre-launch	−0.0069 (0.0297)	0.0180 (0.0462)	0.0702 (0.0609)	0.0351 (0.0408)	0.0412 (0.0489)	0.0778 (0.0552)
Snapshot	**−0.0699** (0.0247)	**−0.1323** (0.0237)	**−0.1286** (0.0342)	**−0.1422** (0.0265)	**−0.1779** (0.0331)	**−0.1564** (0.0492)
TWAP 60 s	0.0395 (0.0855)	0.0310 (0.0856)	0.0000 (0.1157)	**0.2959** (0.1071)	0.2381 (0.1895)	0.0583 (0.2281)
Panel B. Absolute final-10 s return (bp)						
	No controls			Fixed effects and volatility control		
Snapshot	**0.2721** (0.0612)			**0.1736** (0.0277)		
TWAP 60 s	−0.0783 (0.1246)			0.0066 (0.0459)		
Panel C. Placebo window and other assets, final-10 s share						
	Placebo window	Boundary − placebo	ETH	SOL	XRP	
Pre-launch mean	0.0341	−0.0036	0.0305	0.0303	0.0293	
Snapshot	**−0.0015** (0.0004)	**0.0277** (0.0019)	**0.0037** (0.0005)	**0.0070** (0.0007)	**0.0075** (0.0006)	
TWAP 60 s	−0.0008 (0.0006)	0.0005 (0.0012)	0.0001 (0.0010)	0.0007 (0.0012)	−0.0002 (0.0009)	
Traded-cycle volume, snapshot (shares)			9,100	3,800	2,400	

Notes: BTC five-minute-only boundaries unless stated. Panel A reports total within-regime slopes of 30-, 60- and 120-second post-boundary returns on the final-10 s return, computed by linear combination from the regime-interaction model. Near-the-strike uses the 5 bp proxy at $T-61$ with the rule-matched strike. Panel B reports regime effects on the absolute final-10 s return, with the controls in Table 1, Panel C. Panel C reports effects on the placebo share for a window ending at $T-150$, the boundary-minus-placebo gap, and other assets using their own launch dates. The base is pre-launch. Parentheses contain day-clustered standard errors. Coefficients in bold are statistically significant at the 5% level (p<0.05).

final-ten-second return was 0.27 basis points, or 26 percent, larger under snapshot settlement than before the launch, 0.17 points with the volatility control, and it is within 0.08 points of its pre-launch level under the 60-second rule.

The placebo window that ends 150 s before each boundary has a final-ten-second share 0.15 points lower under snapshot settlement and 0.08 points lower under the 60-second rule, so the boundary-minus-placebo gap is 2.77 points in the snapshot period and 0.05 points after the switch. ETH, SOL and XRP show launch effects of 0.37, 0.70 and 0.75 points, between a seventh and three-tenths of BTC's, and 60-second coefficients of 0.01, 0.07 and −0.02 points. Around 5 June 2026, when Polymarket began locking taker orders during its 250-millisecond taker delay, the twelve weekly coefficients stay within 1.8 points of the reference week and none is significant. Around 17 August, when the delay was cut from 250 to 50 ms, the reference week of 10 to 16 August already fell under the averaged rules, with four 30-second and three 60-second days, and the three weeks after the change are within 0.21 points of it. The higher weeks before it run from 20 July to 9 August, when snapshot settlement still held on all but the last three days. The launch effect sits in the tail of the distribution. The median share rose by 0.12 points and the 90th percentile by 5.3 points, and both are back near their pre-launch values under the 60-second rule.

The footprint is a pattern in Binance trading, not a direct measure of manipulation of the settlement feed. Traders who hedge contract positions on Binance and unwind the hedges at settlement would concentrate volume at the same second. Their price pressure can reverse, as when index-arbitrage positions unwind at futures expiration (Stoll and Whaley, 1987). A digital contract's hedge ratio peaks near its strike, so hedging would be largest in near-the-strike cycles, as option hedges help pull stock prices to strikes on expiration dates (Ni et al.,

2005). The fall at the switch could also reflect trading that moved to other venues that enter the Chainlink feed. Two facts weigh against hedging as the main source. Dai et al. (2026) trace gains of about $8.2 million to wallets trading in pushed cycles, which hedging would not produce, and hedges of an averaged contract would be unwound over its averaging window, yet in the first three weeks of the 60-second rule the final-minute share was within 0.15 points of its pre-launch level (Table 1). Migration to other venues cannot be checked with Binance data.

5. Conclusion

While Polymarket's five-minute contract settled on one price, Binance carried a footprint of trading at the settlement time. The final ten seconds before each boundary held 5.7 percent of window volume instead of 3.1, the flow was one-sided, it concentrated in cycles close to the strike and in cycles with the most money at stake, and the price moves it produced partly reversed. In the first three weeks of the 60-second average each of these signatures disappeared within both moneyness groups and at every boundary type. The paper makes two contributions. It provides the first evaluation of a switch from single-print to time-weighted average settlement in which the contract, its horizon, its reference asset and its oracle provider stay fixed and the reference market is observed at one-second resolution. And it gives the benchmark-design literature a measured case in which a footprint from which Dai et al. (2026) document gains of about $8.2 million disappeared in the four weeks after the switch to averaged settlement, while the liquidity rewards paid alongside it were small against the contract's turnover.

For the design of price-settled contracts, the case shows that a contract need not be lengthened to lose most of its settlement footprint. Dai et al. (2026) name longer horizons as the direct remedy. Polymarket kept the five-minute horizon, averaged the settlement price, and the footprint went. The horizon never changed and the two window lengths produced the same final-ten-second shares, so the case does not show whether a longer horizon or a longer window would do better. Online Appendix D shows that from 5 September, the day after Polymarket raised its taker delay and the last day of the reward programme, a smaller footprint returned, mainly on weekends, when Binance volume is thinnest. Whether a one-minute average will be enough once stakes grow is open. Mackinga et al. (2022) show for on-chain oracles that the cost of moving an average can fall far below what its window suggests.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Variable definitions

All times are UTC. The observation is asset a and five-minute boundary T; the associated window contains the 300 s $[T-300, T)$. Let q_j and u_j be Binance quote volume and taker-buy quote volume in second $T-300+j$, $j=0,\ldots,299$, in USDT. Let C_j be that second's close, O_0 the open at $T-300$, $V=\sum_{j=0}^{299} q_j$, and $v_b=\sum_{j=10b}^{10b+9} q_j$. These are exchange observations, not reconstructions of the Chainlink feed. A basis point is 10^{-4} of a log price change, and $\epsilon=10^{-9}$ guards volume denominators. Shares and indicator means are fractions unless explicitly multiplied by 100 in a figure. Table A1 defines each variable. Online Appendices B to D define their additional measures in the text and table notes.

Table A1
Definitions and construction of empirical variables.

Variable/code name	Definition, timing and implementation
Observation, sample and settlement regimes	
Boundary and sample `asset`, `t_end`, `btype`, `fiveonly`	BTC, ETH, SOL or XRP, paired with its USDT spot market on Binance. `t_end` is the Unix timestamp of T, a multiple of 300 s. `btype` is 1 at minutes :05, :10, :20, :25, :35, :40, :50 and :55, 2 at :15, :30 and :45, and 3 at :00. The sample runs from 1 July 2025 to 4 September 2026, with boundaries from 00:05 on the first day through 23:55 on the last, 124,127 per asset. Online Appendix D extends it to 24 September. `fiveonly` equals one for type 1. The main sample holds the 82,752 BTC type-1 boundaries across 431 days, the boundary-type comparison uses all BTC boundaries, and the other-asset comparison uses the type-1 boundaries of each asset.
Settlement regimes `launch_date`, `pm5_rule`, `regime`, `post_twap`	The first trading date is the first UTC date with positive recorded five-minute volume, 12 February 2026 for BTC and 18 February for ETH, SOL and XRP; earlier boundaries are pre-launch. The rule is parsed from each cycle's Gamma `description` as snapshot (1), 30-second TWAP (2), or 60-second TWAP (3). The last snapshot cycle ends at 00:00 on 7 August; the 30-second rule runs from 00:05 that day to 00:00 on 14 August; the 60-second rule starts at 00:05 on 14 August. A cycle without a parsed rule takes the regime of its date. The BTC main-sample counts are 43,392, 33,792, 1,344 and 4,224. `post_twap` equals one under either averaged rule.
Volume and order-flow outcomes	
Volume shares `share10`, `share60`, `s00`–`s29`	$V_K=\sum_{j=300-K}^{299} q_j$ and `share`$K=V_K/(V+\epsilon)$ for $K=10,30,60$; an even spread gives 1/30, 1/10 and 1/5. The principal outcome is `share10`. Bin shares are $s_b=v_b/(V+\epsilon)$, bin b covering $[T-300+10b, T-290+10b)$; bin 29 equals `share10`. The grouped bins of Table 2 are s_0, $s_{21}+s_{22}+s_{23}$, $s_{24}+s_{25}+s_{26}$ and `share30` − `share10`, covering $[-300,-290)$, $[-90,-60)$, $[-60,-30)$ and $[-30,-10)$ relative to T.
Tenfold spike `spike10`	$\mathbf{1}\{(V_{10}/10)/[(V-V_{10})/290+\epsilon]>10\}$, so the volume rate in the last ten seconds exceeds ten times the rate in the preceding 290 s, equivalent to `share10` > 10/39.
Net-flow ratio `netratio10`	$(\lvert F_{10}\rvert/10)/\{(V-V_{10})/290+\epsilon\}$ with $F_{10}=\sum_{j=290}^{299}(2u_j-q_j)$, the signed net taker-buy flow in the last ten seconds. Offsetting buy and sell flows cancel before the absolute value.
Excess terminal volume `excess10_usd`	$V_{10}-V/30$ per boundary, summed within each UTC date and averaged over the days of a regime for the dollar figures.
Placebo window `plc_share10`, `diff_share10`	The final-ten-second share of the window $[T-450, T-150)$, and `share10` minus it. Missing for the first boundary.

(*continued on next page*)

Table A1 (*continued*).

Variable/code name	Definition, timing and implementation
Prices, moneyness and volatility	
Returns `push_bp`, `abs_push`, `post30_bp`, `post60_bp`, `post120_bp`	$10^4[\log C_{299} - \log C_{289}]$, the close at $T-1$ against the close at $T-11$, and its absolute value. Post-boundary returns run from the close at $T-1$ to the close $h = 30, 60, 120$ s later. Reversal entries are total within-regime slopes from a regression with regime intercepts and regime interactions with `push_bp`.
Moneyness `dist_match61_bp`, `ntm5_61_match`	$10^4[\log C_{239} - \log K]$, the Binance close at $T-61$ against a Binance strike K that follows the settlement rule, measured before the averaging window at the close opens. K is the open O_0 at $T-300$ before the switch and the mean of the 1-second closes over $[T-330, T-300)$ or $[T-360, T-300)$ under the 30- or 60-second rule. Near the strike means $\lvert$`dist_match61_bp`$\rvert \le 5$.
Volatility control `rv_first240`, `ln_rv240`	$\sum_{j=1}^{239} r_j^2$ with $r_j = 10^4(\log C_j - \log C_{j-1})$, in squared basis points, and $\log(1 +$ `rv_first240`$)$. The controlled regressions add `ln_rv240` and hour-of-day and day-of-week fixed effects.
Contract volume, grouping and calendar variables	
Contract volume `pm5_volume`, `pm5_traded`, `vol_ok`	Gamma `volumeNum` of the five-minute market with the same asset and end time. It is reported turnover in shares, each paying one dollar if its outcome occurs, not open interest. A missing field stays missing and a zero stays zero; traded cycles have positive volume. Volume analyses exclude 5–26 March 2026 and missing values, leaving 34,476 launched BTC type-1 boundaries, flagged by `vol_ok`.
Volume quintile `pm5_q`	Stata `xtile` on the `vol_ok` observations pooled across the three launched regimes, computed once. The mean-volume row of Table 3 refers to snapshot observations.
Pushed cycle `pushed`	One if `netratio10` exceeds 2.996, its 90th percentile among snapshot BTC type-1 boundaries, a threshold fixed across regimes and moneyness groups.
Daily outcomes and event weeks `relday`, `relweek`	Daily outcomes are unweighted means over the 192 BTC type-1 boundaries of a UTC date. `relday` is the UTC date minus 7 August 2026 and `relweek`$= \lfloor$`relday`$/7 \rfloor$. The event study keeps the 84 dates of weeks −8 to +3, 12 June–3 September 2026, and omits week −1, 31 July–6 August. The placebo comparisons use the same week definition centred on 5 June, with the 91 dates of weeks −8 to +4, and on 17 August, with the 47 dates of weeks −4 to +2 and a reference week, 10–16 August, of four 30-second and three 60-second dates.
Calendar controls `hour_utc`, `dow`, `date_s`, `month_s`, `aug89`	UTC hour and day of week. Standard errors are clustered by the UTC date of T, and monthly means use the UTC calendar month. `aug89` flags 8 and 9 August 2026 for Table 2, Panel C.

Appendix B. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.econlet.2026.113297.

Data availability

Data will be made available on request.

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